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RESEARCH STATEMENT

I am an applied macroeconomist specializing in central bank communication and textual analysis. My research examines how policymakers use language—particularly linguistic complexity—to manage uncertainty, signal policy intentions, and shape market expectations. I combine natural language processing (NLP), econometrics, and empirical macroeconomic methods to study both external communication with markets and internal communication within policymaking committees. Across my work, I develop interpretable text-based measures and link them to macroeconomic conditions, financial-market behavior, and institutional dynamics.

Central Bank Communication and Linguistic Complexity

Question and contribution: How do central banks trade off clarity (anchoring expectations) against flexibility (preserving optionality), and how does this trade-off respond to economic conditions? I introduce refined measures of linguistic complexity—abstractness, informativeness, and disunity—that capture ambiguity and strategic vagueness beyond traditional readability and volume metrics.

Data and methods: I construct a panel of policy statements from 18 central banks (2000–2024), extract linguistic features using NLP tools, and estimate panel fixed-effects models that control for macroeconomic conditions and institutional heterogeneity.

Key results: Complexity rises sharply during episodes of extremely low growth, with statements becoming harder to read, more information-dense, and less cohesive. When central banks ease policy, language becomes more abstract even after conditioning on growth. Geographic differences are pronounced, whereas linguistic differences are not. Cross-country patterns show higher complexity where authorities adopt consistently hawkish or optimistic tones. Inflation-related statements correlate positively with abstractness (≈ 0.38). A substantial share of statements (5–30%) primarily address household-relevant issues, raising concerns about accessibility for more vulnerable audiences.

Policy implication: These results provide a foundation for a quantitative framework that evaluates how central banks adjust communicative complexity under different macroeconomic conditions to balance clarity and flexibility. They offer empirical evidence that complexity is actively managed as part of communication strategy rather than a passive by-product of economic stress.

Complexity in FOMC Communication: Drivers, Differences, and Market Reactions

Question and contribution: How does linguistic complexity in Federal Reserve communication vary with past financial-market conditions, and how is complexity associated with subsequent

market behavior? This paper documents empirical relationships between market turbulence, communication patterns, and short-run market responses.

Data and methods: Using thousands of speeches from FOMC policymakers from 1986 to 2025, I extract syntactic and structural complexity features with Python-based NLP tools and link them to volatility measures, trading volume, and intraday and daily market behavior using fixed-effects models.

Key results: Equity-market volatility is associated with simpler language, while uncertainty in Treasuries, gold, or volatility-linked assets is associated with more structurally complex communication. Speaker characteristics—academic background, professional experience, institutional role, and topic choice—account for substantial heterogeneity in communication styles. Complexity is correlated with higher volatility and trading volume in the minutes following a speech, but these associations reverse at the daily horizon, where complexity is linked to dampened volatility and lower trading intensity.

Broader value: This work expands the literature beyond tone and sentiment by showing how linguistic complexity co-moves with market conditions and is associated with short-run information-processing dynamics. The results highlight communication as both a reflection of prevailing uncertainty and a correlation of how markets absorb policy-relevant information.

Deliberation and Institutional Decision-Making

Question and contribution: How do internal deliberation dynamics within policymaking committees affect policy choices? We exploit the unusually rich transcript record of the Federal Open Market Committee (FOMC) to move from treating the committee as a single "speaker" to studying member-level interaction and its carry-over effects across meeting rounds.

Data and methods: Using automated NLP classification, I label transcripts from March 1976–December 2018 by policy stance, expressed emotion, and collaboration, producing a longitudinal dataset that links deliberative features to votes, rate proposals, and dissents and permits analysis of within-meeting member behavior.

Key results: Member stances in the economic round predict stances in the policy round. Tighter stances from board members forecast tighter committee stances, while looser stances from presidents forecast looser outcomes. Collaboration intensifies in contractions and high-inflation episodes, consistent with consensus-building under stress. Deliberative composition systematically predicts both rate adjustments and dissent behavior.

Broader value: The classification toolkit generalizes to extract features like uncertainty language or topic-specific trends, enabling future studies of communication dynamics in other policymaking bodies.

Ongoing and Planned Work

Building on my empirical work, I plan to develop identification strategies that move beyond reduced-form associations. One direction is to exploit quasi-experimental variation in communication or market conditions—such as unscheduled speeches, unexpected absences, or exogenous scheduling shocks—to study how linguistic complexity co-moves with economic uncertainty in settings closer to causal inference. I also aim to refine the measurement of central bank communication by separating policy statements into their constituent components—economic assessment, policy justification, and forward guidance. Current analyses treat statements as unified texts, but these sections serve distinct communicative purposes and may exhibit different complexity dynamics. Decomposing statements will allow me to identify which layers drive the clarity–flexibility trade-off, distinguish complexity arising from economic explanation versus strategic ambiguity, and provide a more granular understanding of how central banks adjust language under stress.

A second line of work will expand the scope of communication audiences and contexts. Recent housing-market stress and rising mortgage defaults have made central bank communication particularly salient for households, who may be more vulnerable to complex or abstract language. I plan to study how linguistic complexity evolves during periods of heightened household exposure to monetary policy, using case studies and survey-based evidence to examine how non-expert audiences process complex communication. Finally, I am interested in the role of artificial intelligence as a new intermediary in communication consumption. As AI-based summarization and conversational tools become widespread, understanding how these systems transform, amplify, or attenuate complexity will be essential for evaluating communication effectiveness in modern information environments.